

# Mixed-effects models

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# Models with group-specific effects

- Consider the standard linear model:

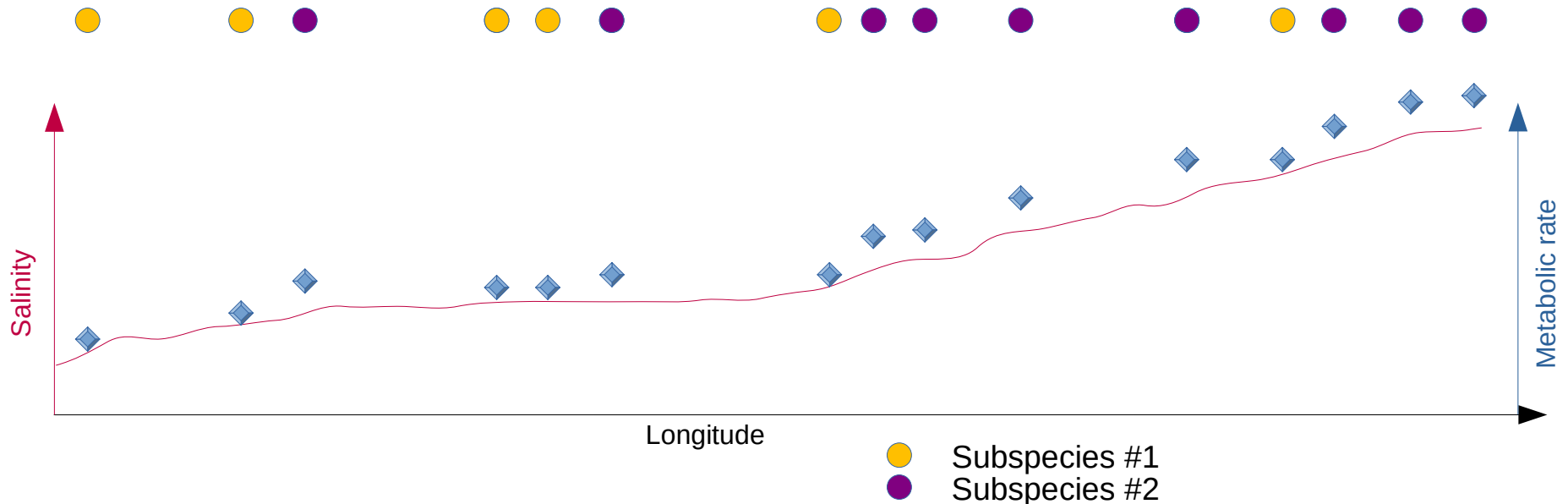
$$f(x) = a + b * x \quad (\text{in R: } \text{lm}(y \sim x))$$

- 

- Consider the case where your samples cover two or more “groups” (e.g. phenotypes, species, ...) (and you’re not particularly interested in them)

# Group-specific effects

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# Group-specific effects

- How can they look like?
  - → Group-specific effect on the intercept
  - → Group-specific effect on the slope
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- $f(x) = (a + a_{g1}) + (b + b_{g2}) * x$       mix 2 groupings, g1 and g2

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This is why it's called mixed: a sum of a fixed (a) and a random ( $a_g$ ) effect

# Mixed models in R

- For linear models: package “lme4”
- `lmer(y ~ (0 + 1 | g) + x)`
- `lmer(y ~ x + (0 + x | g))`
- `lmer(y ~ (0 + 1 | g) + x + (0 + x | g))`
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# Mixed models in R

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> summary(m1)
Linear mixed model fit by REML ['lmerMod']
Formula: y ~ (0 + 1 | g) + x1 + x2 + (0 + x2 | g)
```

REML criterion at convergence: 568.1

Scaled residuals:

| Min      | 1Q       | Median  | 3Q      | Max     |
|----------|----------|---------|---------|---------|
| -2.49190 | -0.68829 | 0.01046 | 0.68436 | 2.58259 |

Random effects:



| Groups   | Name        | Variance | Std.Dev. |
|----------|-------------|----------|----------|
| g        | (Intercept) | 7.2219   | 2.6874   |
| g.1      | x2          | 0.3034   | 0.5509   |
| Residual |             | 1.1680   | 1.0807   |

Number of obs: 180, groups: g, 2

Fixed effects:

|             | Estimate  | Std. Error | t value  |
|-------------|-----------|------------|----------|
| (Intercept) | 4.253620  | 1.916850   | 2.219    |
| x1          | -4.606189 | 0.005188   | -887.848 |
| x2          | 3.372805  | 0.389688   | 8.655    |

Correlation of Fixed Effects:

|    | (Intr) x1    |
|----|--------------|
| x1 | -0.093       |
| x2 | -0.003 0.003 |

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Sums of the fixed and random effects:

```
> coef(m1)
```

```
$g
(Intercept)          x1          x2
a      6.145518 -4.606189  3.762138
b      2.361722 -4.606189  2.983471
```

```
attr(,"class")
[1] "coef.mer"
```

|                      |      |                        |
|----------------------|------|------------------------|
| $(a + a_g)$<br>$g=1$ | $b1$ | $(b2 + b2_g)$<br>$g=1$ |
| $(a + a_g)$<br>$g=2$ | $b1$ | $(b2 + b2_g)$<br>$g=2$ |

# Mixed models in R

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- so if you want to isolate the coefficients of the random effects, you need to take the values on the left and subtract the values from the summary output

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$$a_g + a = 6.14$$

$$a = 4.25$$

$$a_g = 6.14 - 4.25 = 1.89$$

# Mixed models - extended

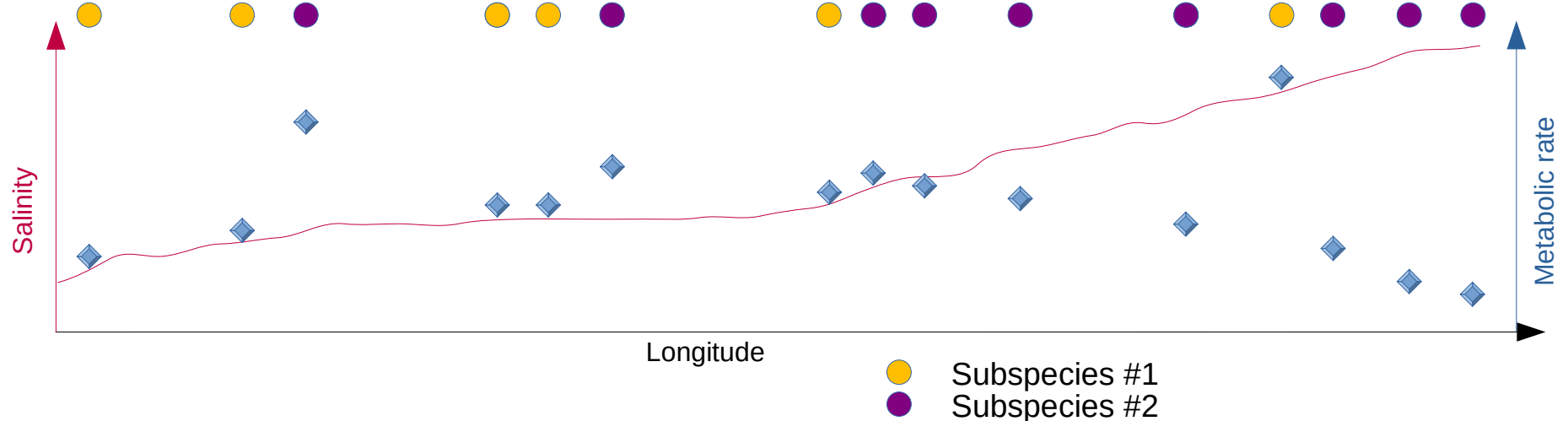
- You can go as complicated as you want...
- → GLMs with mixed effects (package “glmer”)
- → generalized additive models (GAMs) with mixed effects
- → mechanistic models with mixed effects (package “nlmer”)

# Should you care about mixed effects?

- Depends on how much of a change it would make to the fixed effects to include the random effects
- Also depends on prior knowledge (“what happened in the field”?)
- Often, you need relatively many data, because the mixed model is quite complex

# Should you care about M.E.s?

- Imagine this situation:



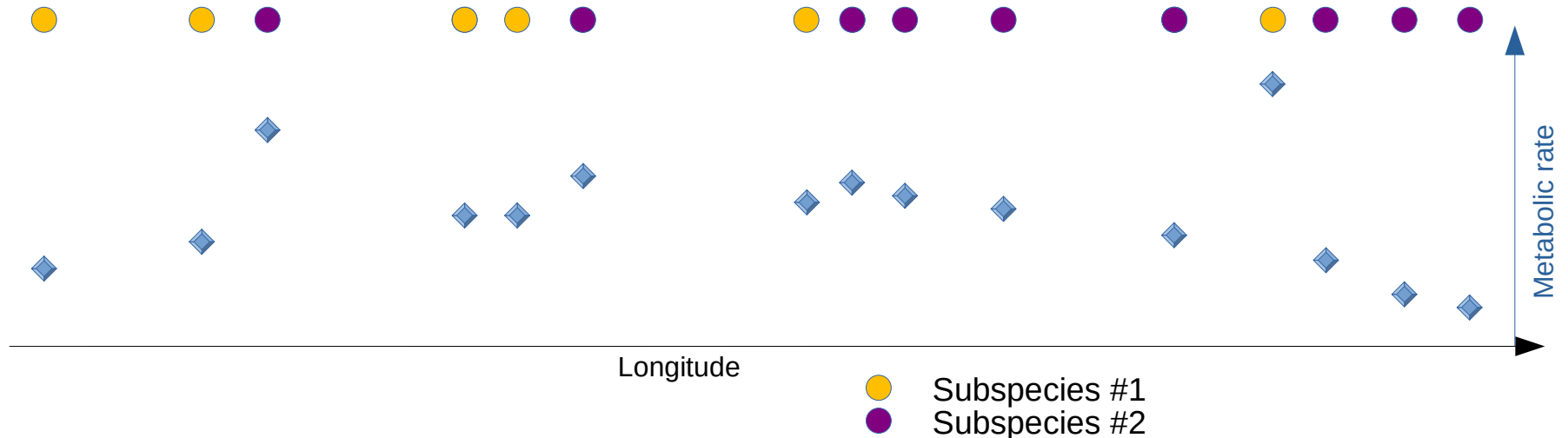


# Should you care about M.E.s?

- Imagine this situation:

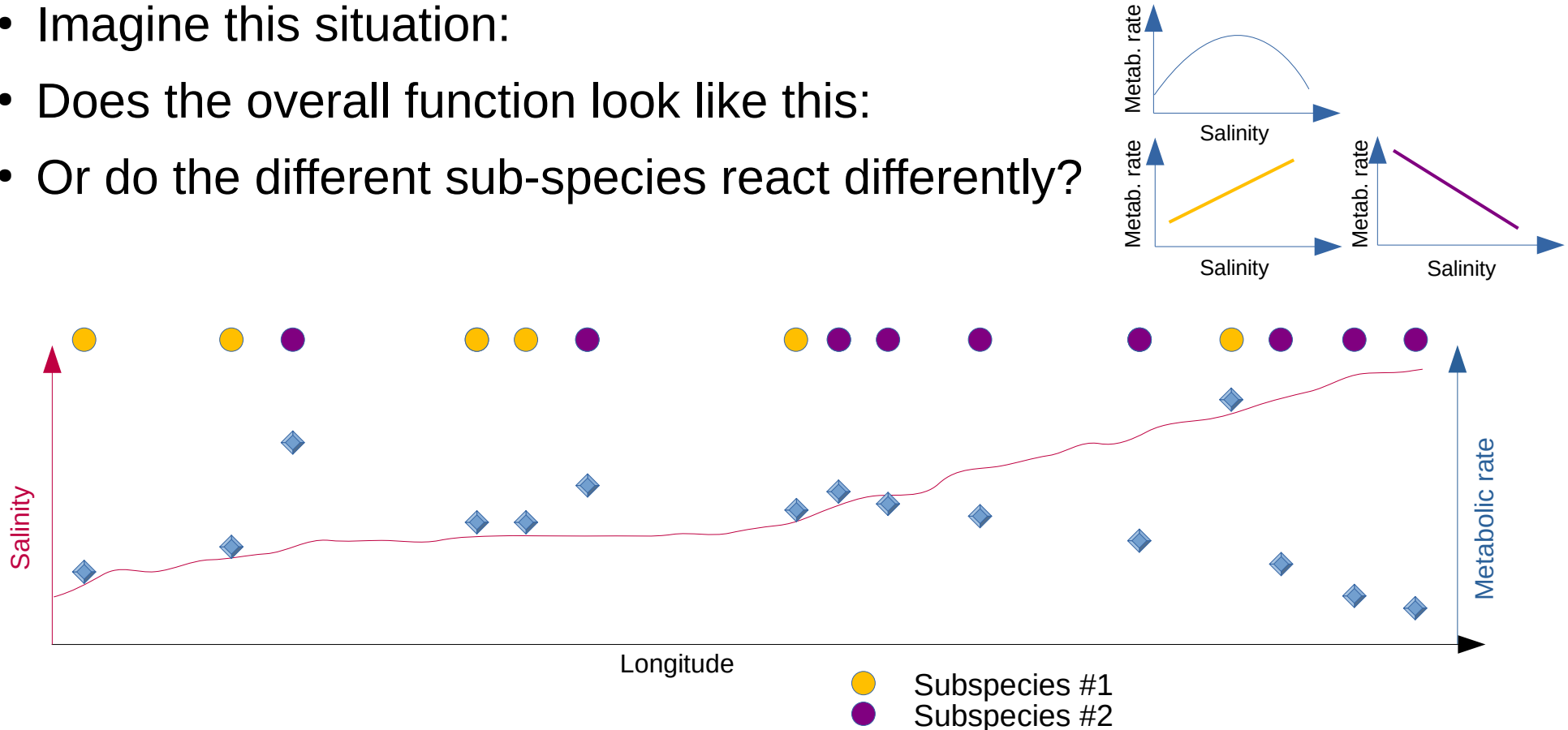
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- 



# Should you care about M.E.s?

- Imagine this situation:
- Does the overall function look like this:
- Or do the different sub-species react differently?



# Should you care about M.E.s?

- How to tell whether a mixed model or a normal (G)LM is the better choice?
- Compare models via diagnostics plots, AIC, sum of squared residuals
- With mixed models you might get better diagnostics plots but higher AIC (because a mixed model usually has more parameters)

# Mixed models and convergence

- With complex (= highly-parameterized) mixed models and relatively few data, you will likely run into this situation:

- 

```
2: In checkConv(attr(opt, "derivs"), opt$par, ctrl = control$checkConv,  :  
   Model failed to converge with max|grad| = 0.00734972 (tol = 0.002, component  
1)
```

- Should you care?
- → Yes, definitely!

# Mixed models and convergence

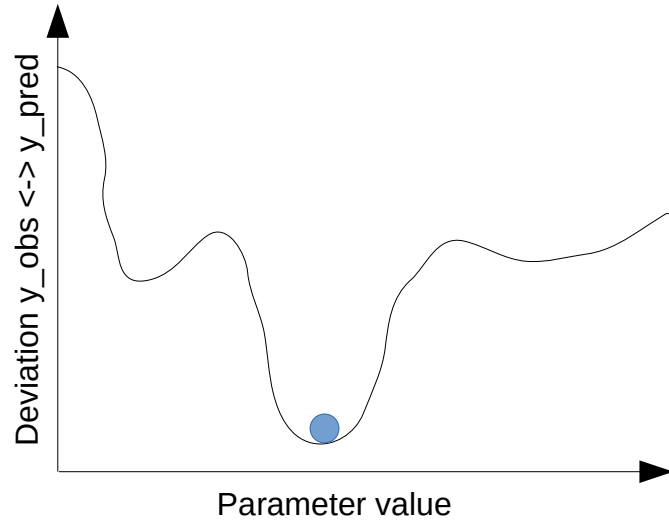
- Why?
- Parameter estimation is not an exact process
- In the background, an algorithm tries to find the “best” parameter values given  $x$ ,  $y$  (the data) and the specified model equation
- “Best” means the least deviation of predicted  $y$  from provided  $y$

# Mixed models and convergence

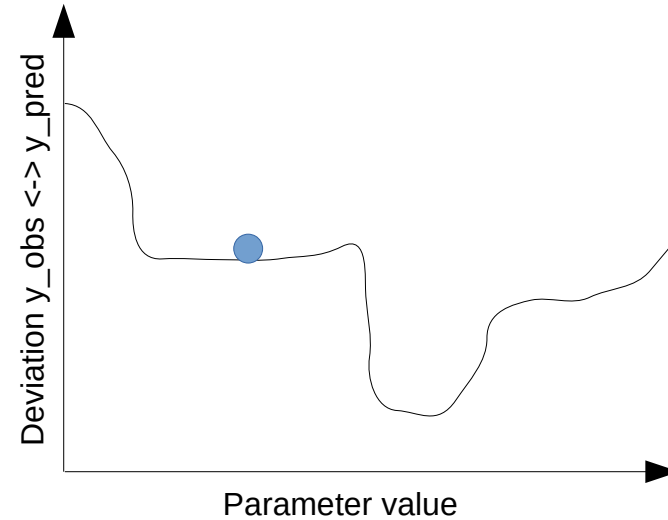
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- Parameter estimation is not an exact process
- In the background, an algorithm tries to find the “best” parameter values given  $x$ ,  $y$  (the data) and the specified model equation
- “Best” means the least deviation of predicted  $y$  from provided  $y$
- Sometimes the algorithm stops without having found the “best” solution. (Very) basically speaking, that is lack of convergence
- → the function didn’t work properly; the parameter estimates returned should not be trusted

# Mixed models and convergence

converged



not converged



# Exercise

- Get together in groups of  $k$  people
- Check if you understood the principle and try to explain to each-other if not (or ask me)
- Create an artificial dataset: make  $x$  values (or  $x_1$  and  $x_2$  values) → then apply an equation (like on slide 6, but you can / should make alterations) → add noise
- Exchange the datasets between groups and try to fit a mixed model → find the underlying relationship



# Exercise #2

- You get a data set
- Try to estimate the underlying model
- It could be a mixed-effects model or a normal linear model
- No data transformation is necessary